# HDOC Day-52

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| **Name** | Gurveer Singh Cheema |
| **SAP ID** | 590032925 |
| **Batch** | 15 |

## Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **arr[]**. Pass the array to a function named **previous_greater()** using its address. Use pointers to pass the array between the calling function and **previous_greater()**.

For each array element, search only the elements to its left and select the nearest element that is greater than the current element. Display **-1** if no greater element exists to the left. Display the results in array order, separated by commas, using a nested-loop approach.

## C Program

```
#include <stdio.h>

void previous_greater(int *arr, int size)
{
    int i, j, found;

    for (i = 0; i < size; i++)
    {
        found = -1;

        for (j = i - 1; j >= 0; j--)
        {
            if (*(arr + j) > *(arr + i))
            {
                found = *(arr + j);
                break;
            }
        }

        printf("%d", found);
        if (i < size - 1)
            printf(", ");
    }
    printf("\n");
}

int main()
{
    int size, i;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", arr + i);

    printf("Previous greater elements: ");
    previous_greater(arr, size);

    return 0;
}
```

## Explanation

The address of **arr** is passed to **previous_greater()** as a pointer. For every element, the inner loop starts from the immediately previous position and moves toward the left. The first value greater than the current element is therefore the nearest previous greater element. If none is found, **-1** is displayed.

## Sample Outputs

```
Example 1:  4 5 2 25     ->  -1, -1, 5, -1
Example 2:  13 7 6 12    ->  -1, 13, 7, 13
Example 3:  1 3 2 4      ->  -1, -1, 3, -1
Example 4:  1 2 3 4 5    ->  -1, -1, -1, -1, -1
```

## Result

Thus, the program successfully finds the nearest previous greater element to the left of each array element using pointers and nested loops.